

Vedant RAJPUT

AI & Machine Learning Engineer · Seeking a 6 month end of studies Internship · Starting : July 2026 · Adaptable to organization's hiring timeline
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PROFILE

Master's student in Data Science & Analytics at EPITA with hands-on experience in **Python**, **deep learning**, end-to-end ML pipelines, and **AWS cloud services**. Published research and applied projects spanning predictive modelling. Experienced with **RAG-based systems**, **LLMs**, and **NLP**. Seeking a 6-month AI & Machine Learning internship to apply research-driven analytical skills to production-scale data challenges.

EDUCATION

Master of Science in Data Science and Analytics, 10/2025 – 03/2027 | Paris, France
EPITA: L'école des ingénieurs en informatique
Bachelor of Technology in Computer Science and Engineering (Équivalent: Master 1 / Bac+4), 09/2020 – 06/2024 | Hyderabad, India
Jawaharlal Nehru Technological University - JNTUH

PROFESSIONAL EXPERIENCE

MetaTaarak AI Innovations Pvt. Ltd | AI Engineer Intern 01/2026 – 04/2026
Structured AI engineering (virtual) programme · AICTE-accredited, under India's Ministry of Education

- Designed and trained **deep learning models** using **TensorFlow**, **Keras**, and **PyTorch** across **supervised** and **unsupervised** learning paradigms, applying regularization, dropout, and hyperparameter tuning to **optimize model performance**.
- Applied end-to-end machine learning workflows** with **Scikit-learn**, **Pandas**, and **NumPy** covering EDA, **feature engineering**, model selection, cross-validation, performance evaluation and **Executed 4+ AI engineering projects** through a structured industry curriculum.

Jawaharlal Nehru Technological University (JNTUH) | Research Assistant Intern 10/2023 – 01/2024

- Conducted faculty-supervised research on **hybrid deep learning** architectures for **medical image classification**, achieving **97%** multi-class classification accuracy (**+7pp over VGG16 baseline**) on **MRI brain tumour datasets**.
- Established reproducible training and **evaluation pipelines** with rigorous **data augmentation**, **transfer learning**, and **cross-validation protocols**, ensuring experimental integrity for peer-review submission. **Validating the hypothesis** that **generative distribution** modelling can enhance discriminative **CNN** feature representations in low-data clinical imaging settings.

PROJECTS

NASA's Helio-Forecast: Geomagnetic Storm Prediction via Deep Learning  01/2026 – 02/2026

- Architected a Bidirectional LSTM (TensorFlow/Keras)** for **geomagnetic storm prediction** 3 hours in advance, achieving an **R² of 0.23** on held-out test data competitive with **published operational baselines (0.15–0.28)** for 3-hour Kp-index forecasting, a domain where the **inherent unpredictability** of solar wind fundamentally constrains **forecast precision**.
- To support this, built a custom **ETL pipeline** to **ingest and impute 24+ months of NASA OMNIWeb satellite time-series telemetry** into a reproducible **training pipeline**, ensuring **data integrity** from **raw sensor ingestion** through to model evaluation.
- Engineered physics-informed features (Eyring Energy: $V \times B_z$)** instead of **raw sensor inputs**, improving the model's capacity to capture magnetospheric energy transfer by 15% – a domain-driven approach validated against baseline LSTM.

European AI Regulatory Compliance Engine (Multi-Agent RAG)  03/2026 – 04/2026

- Engineered a hybrid retrieval pipeline** combining **Qdrant vector search (gemini-embedding-001, 768-dim, COSINE)** and **BM25 keyword search** fused with **Reciprocal Rank Fusion (RRF k=60)**, with **LLM listwise re-ranking at temperature = 0** over **16 candidates** to return **top-8** chunks.
- Built a 3-agent agentic pipeline** (Internal Researcher → External Fact-Checker via DuckDuckGo/FastMCP → Synthesizer). **Designed Sequence Matcher** word-level **chunk** to PDF alignment with highlighted annotations.
- Indexed 280 pages / 1,045,540 characters** across 3 regulatory PDFs (**EU AI Act, GDPR, NIST AI RMF**) into **2,758 semantic chunks** (500-char window, 100-char overlap), achieving **100% generation accuracy** on a hand-labeled 20-question evaluation dataset.

End-to-End MLOps Pipeline: Taxi Fare Prediction  02/2026 – Present

- Architected** a production-grade NYC Taxi Fare prediction system, automating the lifecycle from **data ingestion (Airflow 3)** to model tracking (**MLflow**) and deployment (**FastAPI/Docker**).
- Data Integrity & Governance: Integrated Great Expectations** to enforce data contracts, enforcing **data consistency** and reducing silent failures by validating quality before inference, with all metrics logged via a custom **PostgreSQL/SQLAlchemy** layer.
- Automated Retraining:** Designed a **"Champion-Challenger"** model promotion strategy using **Airflow DAGs** to trigger retraining and evaluation, ensuring the production model remains accurate against live data.

SKILLS

Technical Skills: Python, SQL, PostgreSQL | **AI & Machine Learning:** Hugging Face, Retrieval Augmented Generation(RAG), Large Language Models(LLMs), n8n, CNN, LSTM, Computer Vision, Scikit-Learn, Claude, Gemini, Vector Embedding TensorFlow, Keras, PyTorch, Matplotlib, Seaborn | **Tools & Frameworks:** Git, Docker, AWS, Github Actions (CI/CD), Streamlit, MLflow, Great Expectations, Airflow 3, FastAPI

PUBLICATIONS

Authored and published research paper: 'MRI Brain Tumor Classification using a hybrid VGG16 Nade Model', in the **3rd International Conference on Innovations and Recent Trends in Computer Science (ICIRTCS-23-064)**, December 2023, ISBN: 978-93-91420-60-4.

ACHIEVEMENTS

- Officially Licensed Badminton Player under the Fédération Française de Badminton (FFBAD).
- National Football Player; Captain of College Football (Soccer) team, represented at National, State and University tournaments.

LANGUAGES

English: Bilingual – C2 **French:** Beginner – A2 (under active improvement) **Hindi:** Native – C2 **Marathi:** Native – C2 **Telugu:** Native – C2